



Department of Electrical and Electronic Engineering  
East West University, Dhaka, Bangladesh.

# **EEE 421**

## **RF and Microwave Engineering**

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### Impedance Matching (Stub-tuning)

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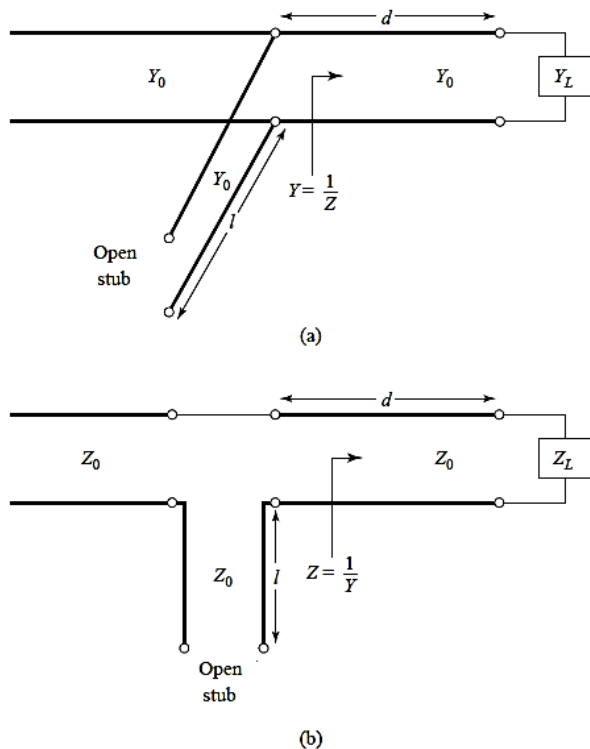
# Stub Tuning Method

## Single-stub Tuning

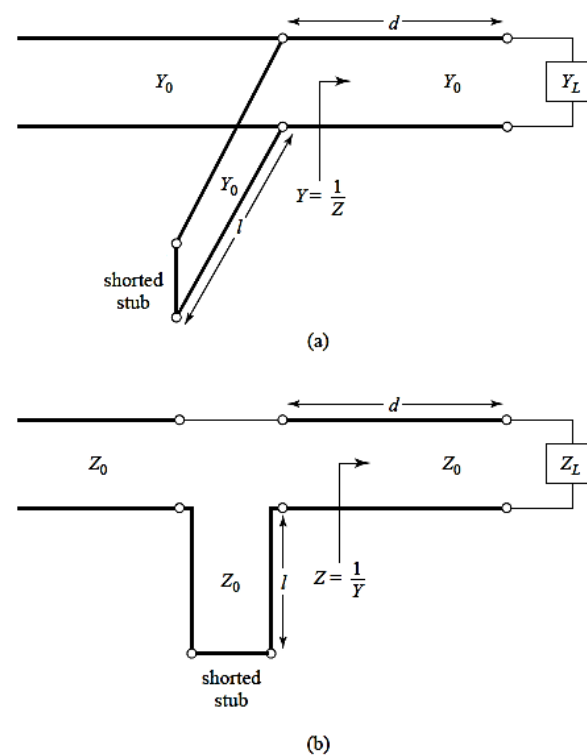
A popular matching technique uses a single open-circuited or short-circuited length of transmission line (a *stub*) connected either in parallel or in series with the transmission feed line at a certain distance from the load

It is often very convenient because the stub can be fabricated as part of the transmission line media of the circuit, and lumped elements are avoided

Shunt stubs are preferred for microstrip line or stripline, while series stubs are preferred for slotline or coplanar waveguide.



Single-stub tuning circuits. (a) Shunt stub. (b) Series stub.



Single-stub tuning circuits. (a) Shunt stub. (b) Series stub.

Advantage:  
able to match any load impedance

Disadvantage:  
Requires a variable length of line  
between the load and the stub

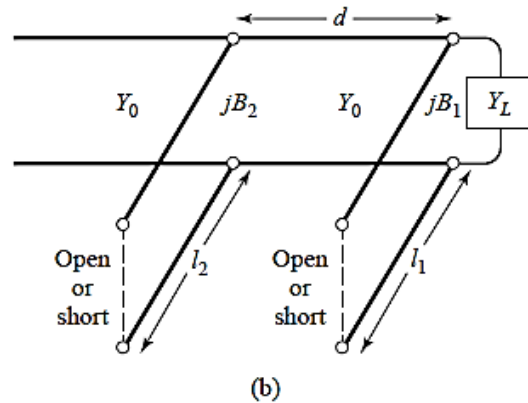
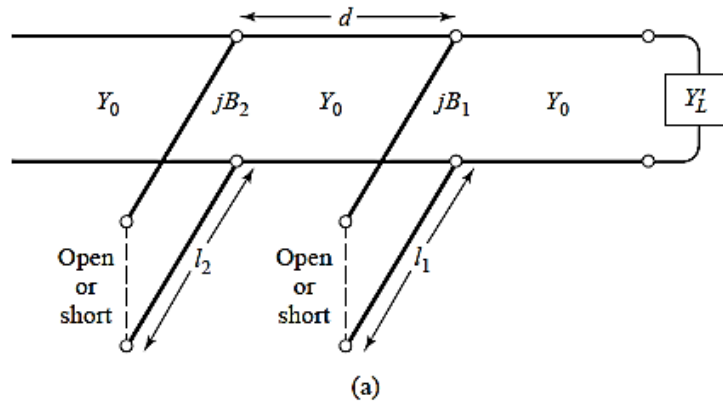
- may not be a problem for a fixed matching circuit,
- would probably pose some difficulty for an adjustable tuner

# Stub Tuning Method

## Double-stub Tuning

Uses two tuning stubs in fixed positions, can be used.

Such tuners are often fabricated in coaxial line with adjustable stubs connected in shunt to the main coaxial line.



Double-stub tuning. (a) Original circuit with the load an arbitrary distance from the first stub. (b) Equivalent circuit with the load transformed to the first stub.

### Advantage:

adjustable stubs can be made for coaxial cables

### Disadvantage:

Cannot match all load impedances.



# Single-Stub Tuning

## EXAMPLE 5.2 SINGLE-STUB SHUNT TUNING

For a load impedance  $Z_L = 60 - j80 \Omega$ , design two single-stub (short circuit) shunt tuning networks to match this load to a  $50 \Omega$  line. Assuming that the load is matched at 2 GHz and that the load consists of a resistor and capacitor in series, plot the reflection coefficient magnitude from 1 to 3 GHz for each solution.

### Solution

The first step is to plot the normalized load impedance  $z_L = 1.2 - j1.6$ , construct the appropriate SWR circle, and convert to the load admittance,  $y_L$ , as shown on the Smith chart in Figure 5.5a. For the remaining steps we consider the Smith chart as an admittance chart. Notice that the SWR circle intersects the  $1 + jb$  circle at two points, denoted as  $y_1$  and  $y_2$  in Figure 5.5a. Thus the distance  $d$  from the load to the stub is given by either of these two intersections. Reading the WTG scale, we obtain

$$\begin{aligned} d_1 &= 0.176 - 0.065 = 0.110\lambda, \\ d_2 &= 0.325 - 0.065 = 0.260\lambda. \end{aligned}$$

Actually, there is an infinite number of distances  $d$  around the SWR circle that intersect the  $1 + jb$  circle. Usually it is desired to keep the matching stub as close as possible to the load to improve the bandwidth of the match and to reduce losses caused by a possibly large standing wave ratio on the line between the stub and the load.

At the two intersection points, the normalized admittances are

$$\begin{aligned} y_1 &= 1.00 + j1.47, \\ y_2 &= 1.00 - j1.47. \end{aligned}$$

Thus, the first tuning solution requires a stub with a susceptance of  $-j1.47$ . The length of a short-circuited stub that gives this susceptance can be found on the Smith chart by starting at  $y = \infty$  (the short circuit) and moving along the outer edge of the chart ( $g = 0$ ) toward the generator to the  $-j1.47$  point. The stub length is then

$$\ell_1 = 0.095\lambda.$$

Similarly, the required short-circuit stub length for the second solution is

$$\ell_2 = 0.405\lambda.$$

This completes the two tuner designs.

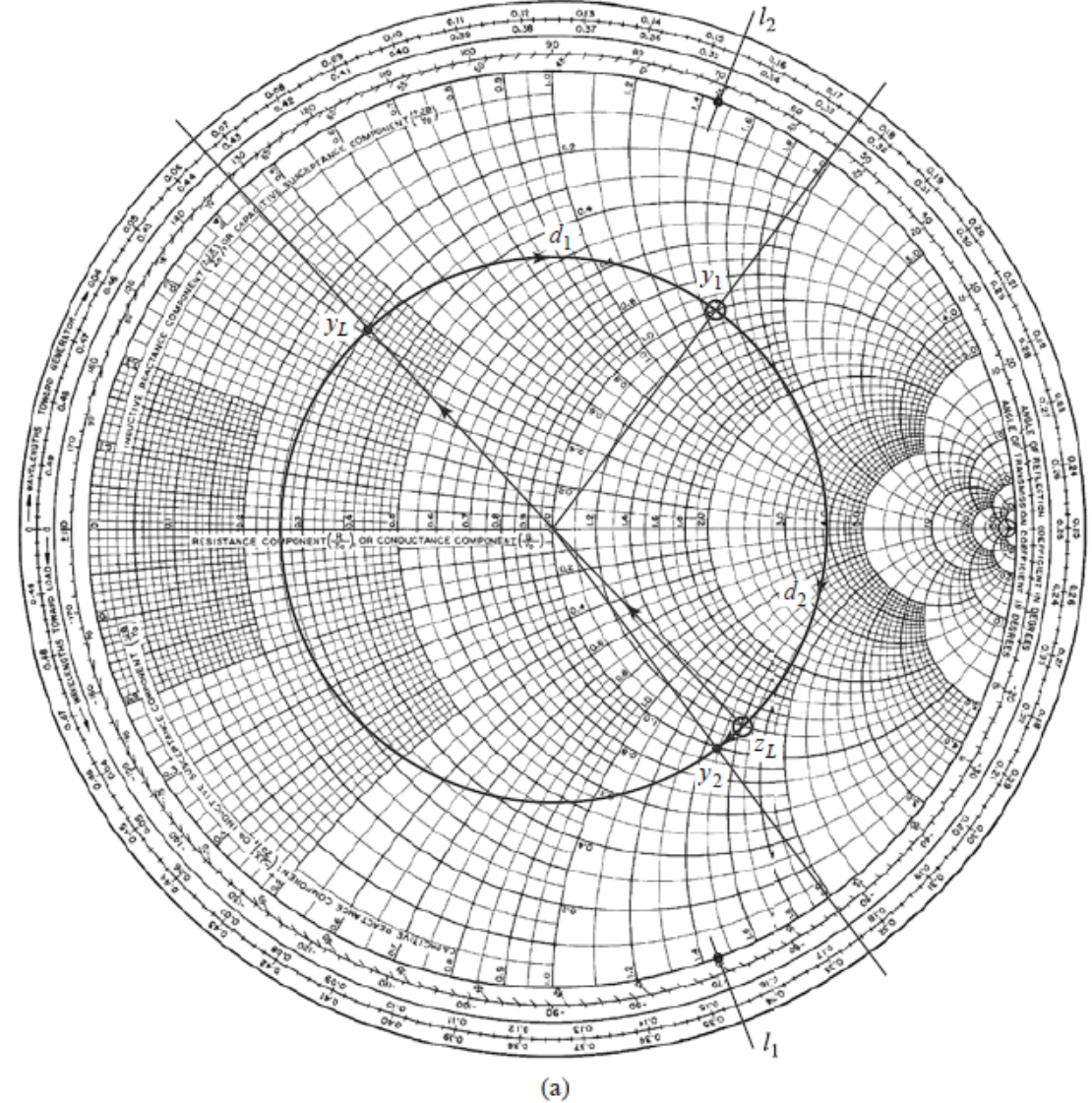


FIGURE 5.5 Solution to Example 5.2. (a) Smith chart for the shunt-stub tuners.

# Single-Stub Tuning

## EXAMPLE 5.3 SINGLE-STUB SERIES TUNING

Match a load impedance of  $Z_L = 100 + j80$  to a  $50\ \Omega$  line using a single series open-circuit stub. Assuming that the load is matched at 2 GHz and that the load consists of a resistor and inductor in series, plot the reflection coefficient magnitude from 1 to 3 GHz.

First plot the normalized load impedance,  $z_L = 2 + j1.6$ , and draw the SWR circle. For the series-stub design the chart is an impedance chart. Note that the SWR circle intersects the  $1 + jx$  circle at two points, denoted as  $z_1$  and  $z_2$  in Figure 5.6a. The shortest distance,  $d_1$ , from the load to the stub is, from the WTG scale,

$$d_1 = 0.328 - 0.208 = 0.120\lambda,$$

and the second distance is

$$d_2 = (0.5 - 0.208) + 0.172 = 0.463\lambda.$$

As in the shunt-stub case, additional rotations around the SWR circle lead to additional solutions, but these are usually not of practical interest.

The normalized impedances at the two intersection points are

$$z_1 = 1 - j1.33,$$

$$z_2 = 1 + j1.33.$$

Thus, the first solution requires a stub with a reactance of  $j1.33$ . The length of an open-circuited stub that gives this reactance can be found on the Smith chart by starting at  $z = \infty$  (open circuit), and moving along the outer edge of the chart ( $r = 0$ ) toward the generator to the  $j1.33$  point. This gives a stub length of

$$\ell_1 = 0.397\lambda.$$

Similarly, the required open-circuited stub length for the second solution is

$$\ell_2 = 0.103\lambda.$$

This completes the tuner designs.

